

GenreMaster: Genre-Conditioned Audio Enhancement with Differentiable DSP

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Abstract—Audio mastering is inherently style dependent: the same spectral, dynamic, and loudness decisions that improve one genre may be inappropriate for another. Existing neural audio enhancement systems rarely model this dependency explicitly, and the lack of large paired mastering datasets makes direct supervised automatic mastering difficult to study. This paper presents GenreMaster, a genre-conditioned audio enhancement framework that combines Transformer-based spectral encoding, feature-wise linear modulation, learnable genre embeddings, and differentiable digital signal processing. Rather than generating audio directly, the model predicts interpretable processing parameters for a mastering-inspired effect chain. We evaluate the framework on GTZAN using a controlled loudness-normalization protocol as a proof-of-concept setting. The results show that genre-conditioned transformations can be learned end-to-end and that differentiable DSP provides a useful bridge between neural optimization and audio-engineering structure. The study does not claim production-ready mastering; instead, it identifies a practical path toward genre-aware automatic mastering once suitable paired datasets and perceptual evaluations become available.

Index Terms—audio enhancement, automatic mastering, genre conditioning, differentiable digital signal processing, FiLM conditioning, music information retrieval

I. INTRODUCTION

AUDIO mastering is the final stage of music production, where a mix is prepared for release through adjustments to loudness, spectral balance, dynamics, and stereo image. These decisions are rarely neutral. A hip-hop master may emphasize low-frequency weight and controlled impact, a classical master may preserve transient detail and dynamic range, and a rock master may favor density without sacrificing intelligibility. As a result, mastering is not only a signal enhancement problem; it is also a style-conditioned decision process. Recent progress in neural audio processing has produced strong results in enhancement, separation, restoration, and production-oriented modeling [1], [2]. However, most systems treat enhancement as a largely genre-agnostic transformation. This assumption is limiting for music production, where the desired target is shaped by genre conventions, listener expectations, and production aesthetics. A system that optimizes average reconstruction quality may still fail to make style-appropriate processing choices.

Developing automatic mastering systems also faces a data bottleneck. Supervised mastering would ideally require paired examples of unmastered and professionally mastered audio, together with genre or style annotations. Such data is difficult to obtain at scale because professional masters are proprietary

and unmastered sources are seldom released alongside final versions. Consequently, research in this area must distinguish between technical feasibility and production readiness. A controlled experiment can test whether a model can learn genre-dependent transformations, but it cannot by itself establish that the model performs professional mastering.

This paper investigates genre-conditioned neural audio enhancement as a step toward automatic mastering. We introduce GenreMaster, a framework that encodes audio features with a Transformer, represents genre labels through learned embeddings, injects those embeddings using feature-wise linear modulation (FiLM), and renders the output through a differentiable digital signal processing (DSP) chain. The model therefore learns to predict parameters for interpretable audio operations rather than directly synthesizing a waveform.

To evaluate GenreMaster in a proof-of-concept setting, we utilize the GTZAN dataset to bypass the lack of large, paired mastering datasets. We construct synthetic input-target pairs by normalizing each clip to a fixed integrated loudness target of -18 LUFS and using the original clip as the reconstruction target. The model therefore learns to recover the original loudness level from a loudness-shifted input. This intentionally limited setup tests whether the proposed conditioning and differentiable processing architecture can learn controlled, genre-dependent transformations, rather than attempting to replace a professional mastering engineer.

The central question addressed in this paper is: can genre information improve a neural audio enhancement pipeline when the processing itself is constrained to an interpretable differentiable DSP chain?

The main contributions of this paper are as follows:

- We present a genre-conditioned audio enhancement framework that combines Transformer spectral encoding, learned genre embeddings, FiLM conditioning, and differentiable DSP.
- We formulate mastering-inspired enhancement as interpretable parameter prediction rather than direct waveform generation.
- We evaluate the framework in a controlled GTZAN proof-of-concept setting using a loudness-normalization protocol with stratified genre splits.
- We analyze the gap between controlled reconstruction performance and production-ready automatic mastering, identifying dataset, evaluation, and style-representation requirements for future work.

II. RELATED WORK

Automatic mastering aims to approximate the final production decisions normally made by an audio engineer. Commer-

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This manuscript reports implementation and experiment results from the GenreMaster project.

cial services such as LANDR and eMastered indicate practical demand, but academic progress remains constrained by the scarcity of paired unmastered and mastered material. Existing research has therefore tended to approach neighboring problems, including source separation, speech and music enhancement, black-box effect modeling, and automatic mixing [3]–[6]. These areas provide useful signal-processing and learning techniques, but they do not fully address the mastering problem. In particular, most prior systems optimize enhancement or reconstruction quality without explicitly modeling the style-dependent choices that determine whether a processed track is appropriate for a given musical context.

Genre classification has long served as a central task in music information retrieval. GTZAN remains a commonly used benchmark for genre analysis despite known limitations such as fixed-length clips, coarse labels, and dataset artifacts [7]. Modern convolutional and residual encoders can learn discriminative genre representations from spectrograms, making genre labels useful control variables for downstream music-processing systems [8], [9]. However, genre labels are coarse style descriptors. They capture broad conventions, but they cannot fully represent artist identity, production era, arrangement density, recording practice, or target aesthetic. Prior genre-aware systems therefore provide a useful starting point for conditioning, while also highlighting the need to avoid treating genre labels as complete descriptions of musical style.

Feature-wise linear modulation provides a simple mechanism for injecting conditioning information into neural networks [10]. By learning scale and bias transformations from a control variable, FiLM can adapt intermediate representations without replacing the main encoder. Differentiable DSP offers a complementary direction. Instead of relying only on unconstrained waveform generation, differentiable audio effects allow gradient-based optimization over physically meaningful parameters [11], [12]. This is especially relevant for production tasks, where equalization, compression, limiting, and related effects have established engineering interpretations. Multi-resolution spectral losses further support perceptually relevant optimization in audio generation and enhancement [13], [14].

Taken together, prior work leaves three gaps unaddressed. Mastering-oriented enhancement has received less attention than neighboring tasks such as source separation and automatic mixing, and the style-dependent decisions specific to mastering have not been directly studied. Genre labels have been used as conditioning signals, but a single genre label cannot distinguish tracks that differ in loudness target, spectral balance, dynamic range, or recording era, and no prior mastering system has examined how much this coarseness limits what a model can learn. Finally, FiLM conditioning and differentiable DSP have each been explored independently, but their combination for genre-aware enhancement remains an open question. The present study addresses these gaps by evaluating style-conditioned enhancement under a controlled protocol, using genre information as an explicit conditioning signal while keeping the learned transformation grounded in differentiable audio-processing operations.

III. METHODOLOGY

A. Problem Formulation

Let x denote an input audio signal and let g denote its genre label. The objective is to learn a transformation f_θ that produces an enhanced signal $\hat{y}$:

$$\hat{y} = f_\theta(x, g). \quad (1)$$

In GenreMaster, this transformation is decomposed into audio encoding, genre conditioning, DSP-parameter prediction, and differentiable rendering. Rather than modeling $\hat{y}$ as an unconstrained neural waveform, the system predicts a parameter vector ϕ for a mastering-inspired processing chain:

$$\phi = P_\theta(E_\theta(x), z_g), \quad (2)$$

$$\hat{y} = D_\phi(x), \quad (3)$$

where E_θ is the audio encoder, z_g is the learned genre embedding, P_θ is the FiLM-conditioned parameter predictor, and D_ϕ is the differentiable DSP chain.

The parameter vector is structured around the mastering controls predicted by the model:

$$\phi = [\phi_{\text{lufs}}, \phi_{\text{eq}}, \phi_{\text{comp}}, \phi_{\text{lim}}, \phi_{\text{width}}], \quad (4)$$

where the terms denote target loudness, equalization, multiband compression, limiter threshold, and stereo-width control, respectively.

This formulation is designed to preserve a connection between the learned model and audio-engineering practice. The network learns how strongly to apply processing, while the rendering stage remains structured around recognizable audio operations.

B. Architecture

The architecture consists of four components, shown in Fig. 1. First, log-mel spectral features are extracted from the input audio and passed to a Transformer encoder [15]. The encoder models temporal dependencies in the spectral representation and produces a compact audio feature vector.

Second, each integer genre label indexes into a learnable weight matrix of shape 10×256 , retrieving a 256-dimensional dense vector that is optimized jointly with the rest of the model. Third, the genre embedding modulates the audio representation using FiLM:

$$\text{FiLM}(h, z_g) = \gamma(z_g) \odot h + \beta(z_g), \quad (5)$$

where h is the encoded audio representation and $\gamma(\cdot)$ and $\beta(\cdot)$ are small feed-forward networks that take the genre embedding as input and output a per-feature scale and bias, respectively.

Finally, the conditioned representation is mapped to parameters for a differentiable DSP chain. The chain includes loudness control, parametric equalization, multiband compression, limiting, and stereo-width adjustment. These modules are selected because they correspond to common mastering operations while their parameters can be optimized by gradient descent. It is worth noting that the model predicts a single global parameter set per clip rather than time-varying parameters. Dynamic effects such as compression and limiting

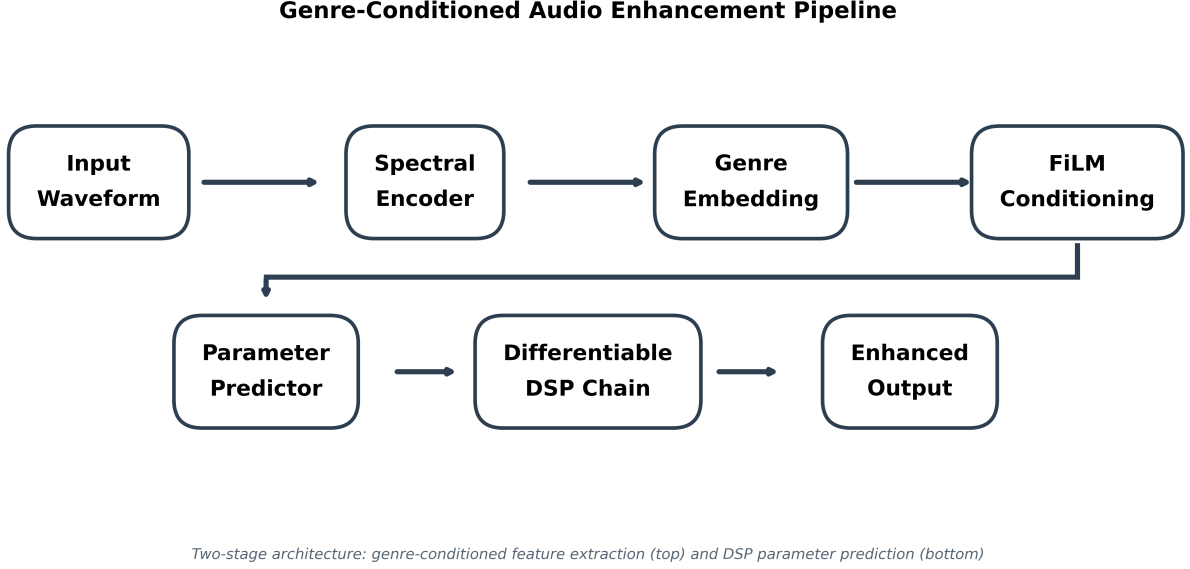


Fig. 1. GenreMaster pipeline. Audio features are encoded, modulated by learned genre embeddings, and mapped to parameters for a differentiable mastering-inspired DSP chain.

are therefore applied with fixed thresholds and ratios across the full signal, whereas in professional mastering these settings often change across song sections. This is a deliberate simplification of the proof-of-concept setting and a clear direction for future work.

Training minimizes a weighted combination of loss terms,

$$\mathcal{L}_{\text{total}} = \lambda_{\text{loud}} \mathcal{L}_{\text{loud}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}}. \quad (6)$$

Two components are particularly important for mastering-

each targeting a different subset of the parameter vector ϕ : the loudness term supervises ϕ_{lufs} , the spectral term supervises ϕ_{eq} , the dynamic term supervises ϕ_{comp} and ϕ_{lim} , and the perceptual term provides a holistic penalty over the full output:

oriented behavior. The loudness term penalizes differences in global level using an RMS-based decibel approximation:

$$\mathcal{L}_{\text{loud}} = \frac{1}{B} \sum_{i=1}^B |20 \log_{10}(\text{RMS}(\hat{y}_i) + \epsilon) - 20 \log_{10}(\text{RMS}(y_i) + \epsilon)|, \quad (7)$$

where B is the batch size and ϵ is a small constant for numerical stability. The dynamic term penalizes differences in

crest factor so that peak-to-average behavior is also matched:

$$\mathcal{L}_{\text{dyn}} = \frac{1}{B} \sum_{i=1}^B \left| 20 \log_{10} \frac{p(\hat{y}_i) + \epsilon}{\text{RMS}(\hat{y}_i) + \epsilon} - 20 \log_{10} \frac{p(y_i) + \epsilon}{\text{RMS}(y_i) + \epsilon} \right|, \quad (8)$$

where $p(\cdot)$ denotes a differentiable peak approximation. The loss design reflects the fact that mastering quality cannot be captured by a single waveform distance. Loudness alignment, spectral balance, dynamic behavior, and perceptual similarity each capture a different aspect of the desired transformation. In

the present proof-of-concept, the loss is used to test learnability under controlled conditions rather than to define a complete perceptual mastering criterion.

C. Data and Experimental Setup

The experiment uses GTZAN after removing one corrupted file, yielding 999 valid clips across ten genres. The data are split in a stratified manner into training, validation, and test partitions, as summarized in Table I. All audio is re-sampled to 22,050 Hz and represented as 30-second clips. Since GTZAN contains finished music rather than unmastered/mastered pairs, we construct synthetic training pairs using a loudness-normalization protocol. Each clip is normalized to a fixed integrated loudness of -18 LUFS using ITU-R BS.1770 measurement; the normalized clip serves as the model input and the original clip as the reconstruction target. The input therefore differs from the target only in overall gain, not in spectral shape, dynamic range, or any other signal property. This is a deliberate simplification: it provides a tractable, reproducible learning problem that tests whether genre conditioning and differentiable DSP can be trained end-to-end, while making no claim that the task represents the full complexity of professional mastering.

To establish that the GTZAN labels provide a meaningful conditioning signal, we first trained a separate ResNet-based genre classifier on the same corpus. This auxiliary stage was used to verify that genre structure is discriminable in the dataset before running the mastering experiment. The reported Transformer-based mastering model was then trained with genre labels as the conditioning variable, using learned genre embeddings within the FiLM-conditioned parameter-prediction pipeline.

For the mastering model, training was configured for a maximum of 50 epochs with validation monitored after each epoch. Gradient accumulation was used to increase the effective batch size, the learning rate followed a cosine schedule with a short warmup phase, and the best checkpoint was selected according to validation loss. Early stopping was enabled with a patience of 15 epochs without improvement, while intermediate checkpoints were saved periodically for recovery and inspection.

TABLE I
DATASET SPLIT CONFIGURATION

Set	Clips
Training	799
Validation	99
Test	101

IV. RESULTS

Fig. 2 shows the training and validation trajectories. The model converges rapidly and maintains a small gap between training and validation loss. This behavior suggests that the architecture is able to learn the controlled degradation-reversal task without severe overfitting.

A. Test Performance

Table III reports the held-out test results. The test loss closely matches the best validation loss, and waveform-level

TABLE II
TRAINING CONFIGURATION

Component	Value
Random seed	42
Sample rate	22,050 Hz
Clip duration	30.0 s
Batch size	8
Gradient accumulation	4
Effective batch size	32
Maximum epochs	50
Optimizer	AdamW [16]
AdamW betas	(0.9, 0.999)
AdamW epsilon	1×10^{-8}
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
Scheduler	Cosine with warmup
Warmup epochs	2
Gradient clipping	1.0
Checkpoint interval	Every 5 epochs
Model selection	Lowest validation loss
Early stopping	Patience 15
Executed epochs	32
Device backend	CUDA

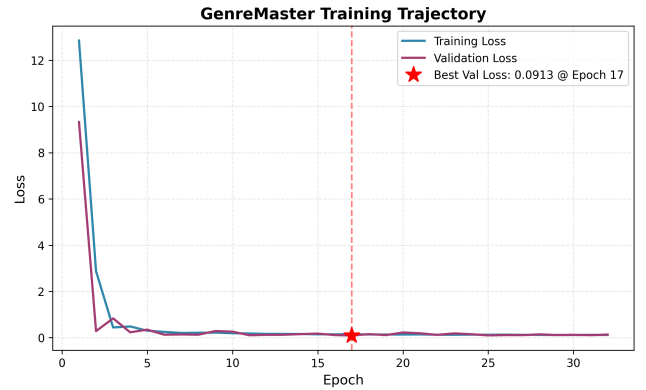


Fig. 2. Training and validation loss curves for GenreMaster on the controlled GTZAN enhancement task.

metrics indicate strong reconstruction quality in the synthetic setting. These values should be interpreted as evidence that the architecture learns the controlled transformation, not as evidence of professional mastering quality.

TABLE III
HELD-OUT TEST PERFORMANCE

Metric	Value
Test samples	101
Validation loss	0.0913
Test loss	0.0903
SNR	16.67 dB
Correlation	0.9892
Best epoch	17

B. Loss Component Analysis

Fig. 3 breaks down the total test loss by component. Loudness and dynamic terms together account for 83.6% of the total loss (48.8% and 34.8% respectively), while the spectral term contributes only 0.3%. This distribution directly reflects

the nature of the task: because the input differs from the target only in overall gain, the spectral content is already well-matched and the model has little spectral error to correct. The dominant loudness and dynamic residuals indicate that even a global gain shift creates measurable differences in RMS-based loudness and crest factor that the model does not fully resolve. The near-zero spectral loss also exposes a limitation: the current protocol does not exercise the equalization component of the DSP chain, so the spectral branch of the model receives almost no training signal.

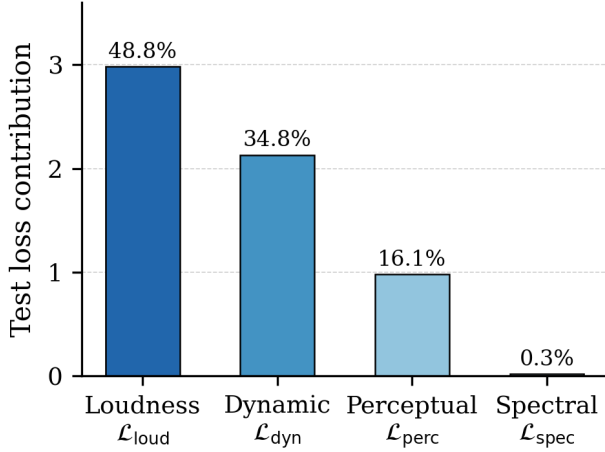


Fig. 3. Test loss broken down by component. Loudness and dynamic terms dominate (83.6% combined), while the spectral term is negligible (0.3%), consistent with a task that differs only in overall gain.

To verify that the dataset supports meaningful genre-conditioned modeling, we first trained a separate ResNet-based genre classifier on the same corpus. The classifier reached a best validation accuracy of 88.9%, as shown in Fig. 4, indicating that genre labels are learnable from the spectral representation and are consistent with published GTZAN baselines. This auxiliary result motivated the subsequent mastering experiment, in which genre labels were used as the conditioning signal for the Transformer-based model. At the same time, GTZAN genre labels remain coarse and should not be treated as complete style descriptors.

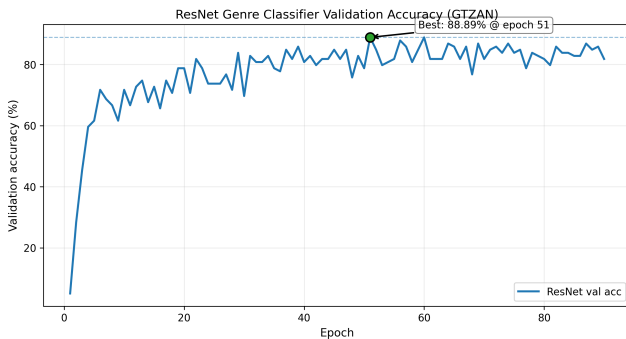


Fig. 4. Validation accuracy trajectory for the auxiliary ResNet genre classifier on GTZAN.

V. DISCUSSION

The experiment tests a specific and narrow claim: can a genre-conditioned differentiable DSP pipeline learn to reconstruct an audio clip from a loudness-shifted version of itself? The answer is yes, within the controlled GTZAN setting. The model learns to apply a corrective gain through the DSP chain, and it does so with a small validation-test gap, indicating that the learned behavior generalizes across the stratified split.

However, this falls well short of genre-conditioned mastering. The task reduces to loudness recovery — the input and target differ only in overall gain, so the model has no incentive to apply genre-specific equalization, compression shaping, or stereo decisions. The loss component analysis confirms this: the spectral term contributes only 0.3% of the total loss, meaning the equalization branch of the DSP chain receives almost no training signal. The genre embedding is present in the architecture, but the current task does not require it to do meaningful work — a genre-agnostic gain correction would achieve similar results.

What the experiment does demonstrate is that the architecture is trainable end-to-end and that the differentiable DSP formulation does not introduce instabilities. The Transformer encoder, FiLM conditioning pathway, and DSP parameter predictor can all be jointly optimized under the adopted protocol. This is a necessary but not sufficient condition for a mastering system. The architecture provides a foundation that could support genre-conditioned processing once a richer task — one that requires spectral, dynamic, and stereo decisions that genuinely differ across genres — is used for training.

The main limitation is the absence of real mastering pairs. GTZAN contains already-finished music, and the degraded inputs are generated synthetically. Real pre-mastering material differs in recording quality, mix balance, headroom, noise, room coloration, arrangement density, and artistic intent. A model trained only to reverse synthetic degradation may learn useful signal transformations without learning the judgment required for professional mastering.

The evaluation metrics are also limited. Reconstruction loss, correlation, and SNR are useful for controlled experiments, but mastering is ultimately perceptual and contextual. A technically close reconstruction is not necessarily a better master. Future evaluations should include loudness and true-peak compliance [17], perceptual audio metrics [18], [19], comparisons with commercial systems, and listening studies involving trained engineers.

Finally, genre labels are an imperfect representation of style. Production conventions vary within genres and evolve over time. A more complete system should learn continuous style embeddings that can capture subgenre, era, artist preference, and reference-track intent.

A. Path Toward Automatic Mastering

The present study suggests three requirements for moving from proof-of-concept enhancement to automatic mastering. First, the field needs paired datasets containing unmastered and professionally mastered versions across diverse genres and production contexts. Second, models should be evaluated

with perceptual and preference-based criteria rather than reconstruction metrics alone. Third, conditioning should move beyond discrete genre labels toward richer style representations that can reflect real mastering goals.

Under these conditions, the GenreMaster architecture could serve as a foundation for a more realistic system: the Transformer encoder would model musical context, the conditioning pathway would represent target style, and the differentiable DSP chain would provide controllable, interpretable rendering.

VI. CONCLUSION

This paper presented GenreMaster, a genre-conditioned audio enhancement framework that combines Transformer spectral encoding, learned genre embeddings, conditioning mechanisms, and differentiable DSP within a mastering-inspired parameter-prediction pipeline. The central idea is to treat automatic mastering not as unconstrained waveform generation, but as structured control of interpretable audio-processing operations. This design keeps the learned transformation connected to established mastering practice while still enabling end-to-end optimization.

Within the controlled GTZAN proof-of-concept setting, the study showed three main results. First, the architecture is trainable end-to-end: the Transformer encoder, FiLM conditioning pathway, and differentiable DSP chain can be jointly optimized without instability. Second, the model generalizes from the training split to held-out test clips, with a small validation-test gap under the loudness-normalization protocol. Third, the loss component analysis reveals that the task effectively reduces to loudness recovery — the spectral loss contributes only 0.3% of the total, meaning the equalization and spectral branches of the DSP chain receive almost no training signal under the current setup.

These findings are encouraging, but they should be interpreted with appropriate restraint. The present experiment demonstrates technical feasibility under synthetic supervision; it does not establish professional-quality automatic mastering. Because GTZAN contains finished music rather than genuine unmastered/mastered pairs, the task remains a controlled reconstruction problem rather than a full mastering benchmark. In addition, the reported metrics capture reconstruction fidelity more directly than perceptual preference, which limits the strength of any claims about listener-facing quality.

The main contribution of this work is therefore methodological. GenreMaster provides a concrete experimental framework for studying how style information, neural sequence modeling, and differentiable audio processing can be combined in a mastering-oriented setting. Future work should extend this foundation with real mastering datasets, richer style representations beyond coarse genre labels, perceptual and engineer-in-the-loop evaluation, and stronger comparisons against commercial and professional baselines. Under those conditions, the proposed framework could evolve from a proof-of-concept enhancement system into a more realistic approach to genre-aware automatic mastering.

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